

# Brian Bertacco

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## Personal Profile

Aerospace Engineering MSc graduate from KTH Royal Institute of Technology in Stockholm, focusing on the Space and Defence sectors, with a deep interest in hypersonic flows and propulsion systems. I am resilient, determined, and always ready to rise to challenges; perseverance is my greatest skill. Building on a strong multi-disciplinary engineering background, I have developed effective teamwork skills through complex projects, discovering a natural aptitude for coordination and leadership. I am currently seeking a position as a Propulsion, Defence, Aerospace, Fluid Dynamics or R&D Engineer to apply my technical expertise, contribute meaningfully to the industry and improve myself.

## Education

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**MSc Aerospace Engineering, KTH Royal Institute of Technology | Stockholm, Sweden | 2024 – 2026 | Grade: 4.69/5 GPA**

- **Master's Thesis:** "Experimental characterization of shock wave transmission through different material interfaces."
- **Key Coursework:** Advanced Compressible Flows, Spacecraft Dynamics, Rocket Propulsion, Applied CFD.

**BSc Aerospace Engineering, University of Pisa | Pisa, Italy | 2019 – 2024 | Grade: 103/110**

- **Bachelor's Thesis:** "Radioisotope Thermoelectric Generators for Space Missions and Radiological Safety."
- **Key Coursework:** Aircraft Propulsion, Aircraft Manufacturing Technologies, Applied Mechanics, Materials Science.

## Technical Skills

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- **Simulation & CFD:** ANSYS Fluent, ANSYS Freeflow, CFX, THERMICA, SYSTEMA, MATLAB (Advanced).
- **Design & CAD & Systems:** SolidWorks, CATIA V5, Fusion 360, AutoCAD, SolidEdge, DOORS.
- **Lab & Experimental:** Shock tube testing, Sensor integration, High-Speed Shadowgraphy, Data acquisition, 3D printing.
- **Languages:** Italian (Native), English (C1/Full Professional), German (A2), Spanish (A2), Swedish (A1).

## Project & Research Experience

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**MSc Thesis Research | KTH | Researcher | January 2026 - June 2026**

- Conducted experimental research on shock wave transmission through material interfaces, utilizing 3D-printed PLA, Aluminum and Copper samples. Research is directly applicable to hypersonic aerostructures and defence technologies.
- Designed and executed shock-wave experiments using a pulsed power generator, exploding foils, high-speed shadowgraphy, and FOPH.
- Developed custom analysis scripts, including advanced coordinate rotation logic, to accurately align and analyze transmitted versus reflected wave data.

**KTH Rocketry Team (ÆSIR) | Stockholm | Propulsion & Structures Engineer | 2024 - 2026**

- **Propulsion Design:** Developed the propulsion system for Mjøltnir, a hybrid paraffin-NOx rocket (16cm OD, 4.5m length) targeting a 10km+ altitude. Conducted detailed CFD analyses to optimize engine performance.
- **Testing & Analysis:** Executed static fire tests and sensor calibration. Analyzed critical load cell data to ensure system safety and validate oxidizer tank loading metrics.

**Airbus Sloshing Rocket Workshop (EUROAVIA) | Sloshing CFD Lead | January 2026 – August 2026**

- **Project Leadership:** Secured 3rd place overall at the European finals in Rzeszów, Poland, following an outstanding 90.1/100 score on the Final Design Report (FDR) and a successful launch of a deployable-wing water rocket.
- **Systems Engineering:** Utilized DOORS for rigorous requirement capture, compliance tracking and final design matrix.
- **CFD Simulation:** Executed transient Volume-of-Fluid (VOF) simulations in ANSYS Fluent to model complex free-surface two-phase sloshing dynamics.
- **Design & Validation:** Engineered a mixed baffle configuration utilizing annular rings and a central PLA cruciform mast, achieving a 72.7% reduction in peak lateral sloshing forces.

**MIST (Miniature Student Satellite) | Stockholm | Thermal Analysis Engineer | 2025 - 2026**

- Performed thermal simulations using SYSTEMA and THERMICA to validate satellite survivability for a 2027 orbital launch.
- Collaborated with a multidisciplinary team to define and optimize thermal control subsystem requirements.

## Other Interests

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- Passionate about hiking and nature I have reached several major mountain peaks such as Punta Giordani at 4046 meters (Monte Rosa), Gran Zebrù at 3857m and Piz Corvatsch at 3450m. Sports are my outlet; I enjoy gym workouts and boxing to relax after long days of studying or working.